

NexCart Platform

System Diagrams Atlas

Architecture, Security, Database, AI Pipelines, and Workflow Diagrams

Document: Standalone diagram reference
Date: July 2026
Total Figures: 22 diagrams
Companion: NexCart_Technical_Report.pdf

Contents

1	System and Deployment Architecture	1
2	Security Architecture	3
3	Database Schema Diagrams	6
4	Concurrency and Transaction Patterns	9
5	Business Process Workflows	11
6	AI and Recommendation Systems	13
7	Frontend Architecture	16

List of Figures

1.1	NexCart Three-Tier Architecture with AI Sidecar	2
1.2	Monorepo Structure and Docker Deployment	2
2.1	NexCart Layered Security Architecture	4
2.2	Express Middleware and Security Pipeline	4
2.3	JWT Authentication and Token Blacklisting Flow	5
2.4	Multi-Layer Rate Limiting Architecture	5
3.1	Database Schema — Identity and Roles Domain	7
3.2	Database Schema — Commerce Domain	7
3.3	Database Schema — Financial and Accounting Domain	7
3.4	Database Schema — Recommendation and AI Attribution Domain	7
3.5	Database Schema — Security and Session Domain	8
4.1	Concurrency Control Patterns	10
5.1	B2C Order Checkout Flow (COD and Prepaid)	12
5.2	B2B RFQ Negotiation Workflow	12
6.1	Hybrid Recommendation Score Fusion Pipeline	14
6.2	Offline Recommendation Job Pipeline	14
6.3	AI Business Advisor RAG Pipeline	14
6.4	AI Khatta Vision OCR Digitisation Flow	15
6.5	Two-Step OTP Registration and Login Security Flow	15
7.1	Frontend Application Architecture	17
7.2	Role-Based Access Control Matrix	17
7.3	Razorpay Webhook Security and Idempotent Processing	17

Chapter 1

System and Deployment Architecture

Figure 1.1: NexCart Three-Tier Architecture with AI Sidecar

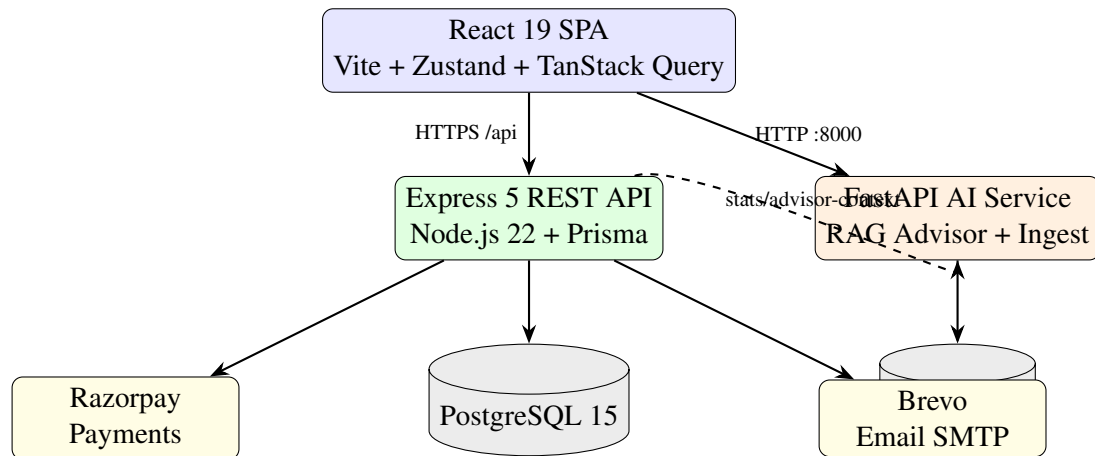
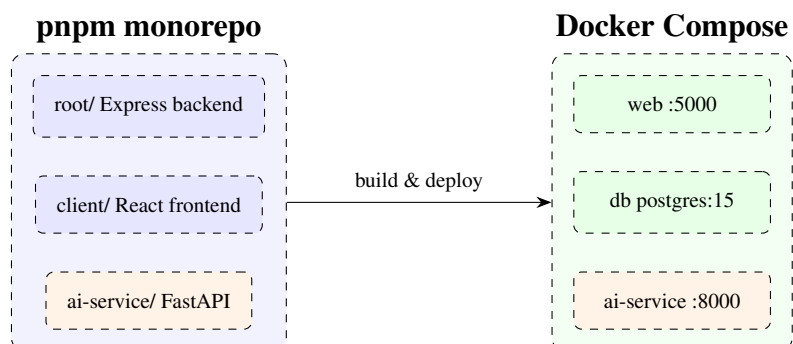


Figure 1.2: Monorepo Structure and Docker Deployment



Chapter 2

Security Architecture

Figure 2.1: NexCart Layered Security Architecture

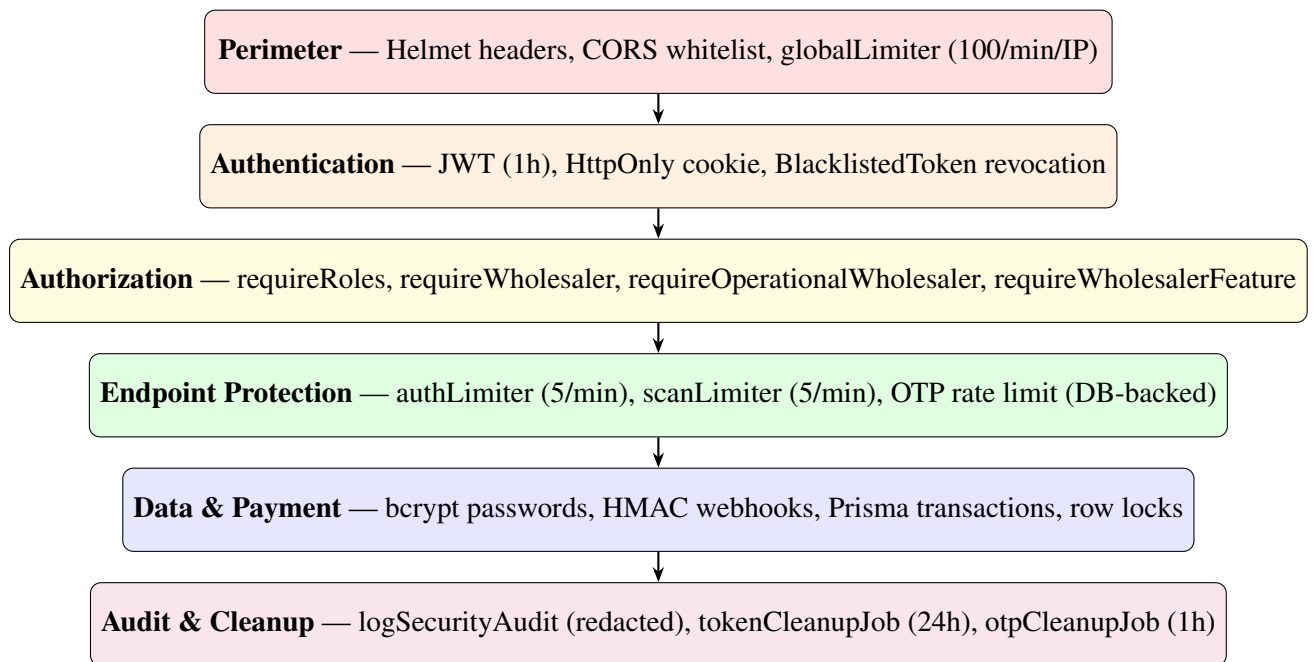


Figure 2.2: Express Middleware and Security Pipeline

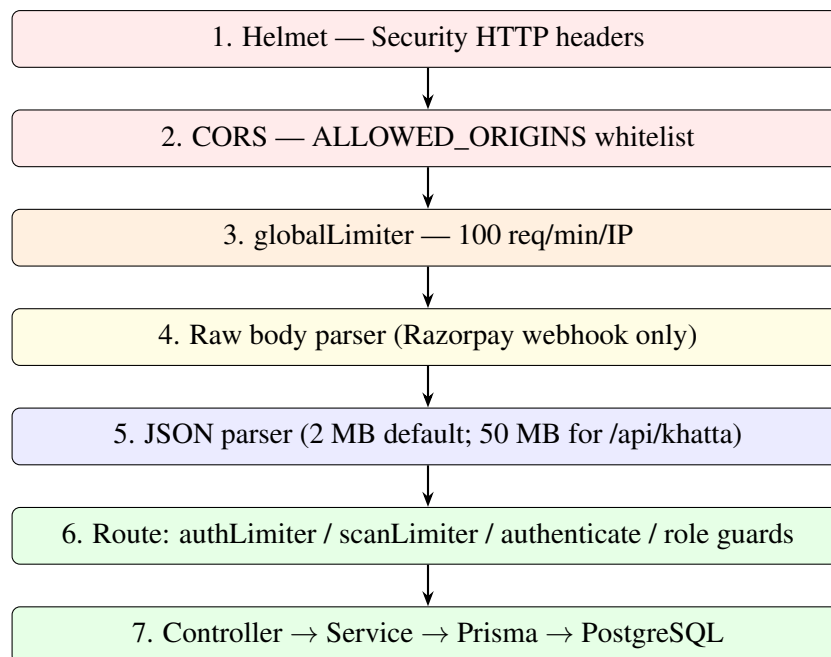


Figure 2.3: JWT Authentication and Token Blacklisting Flow

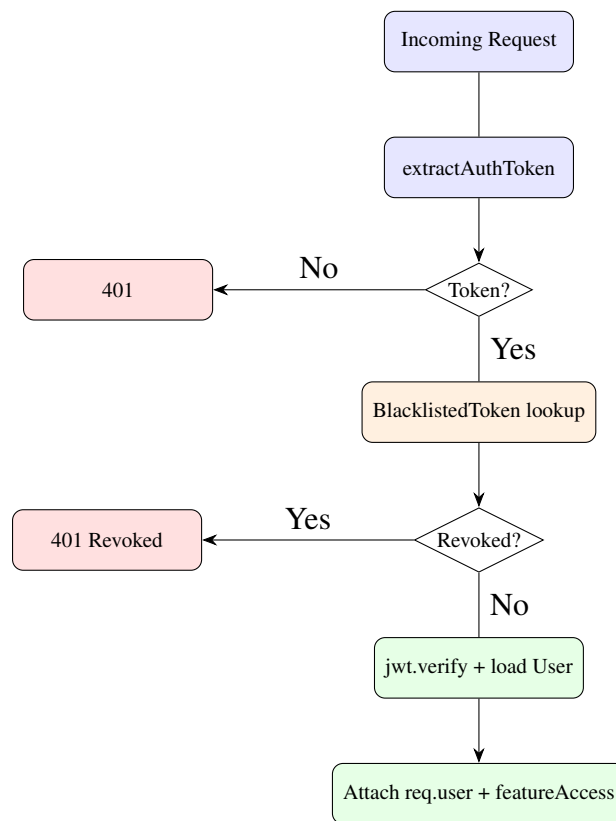
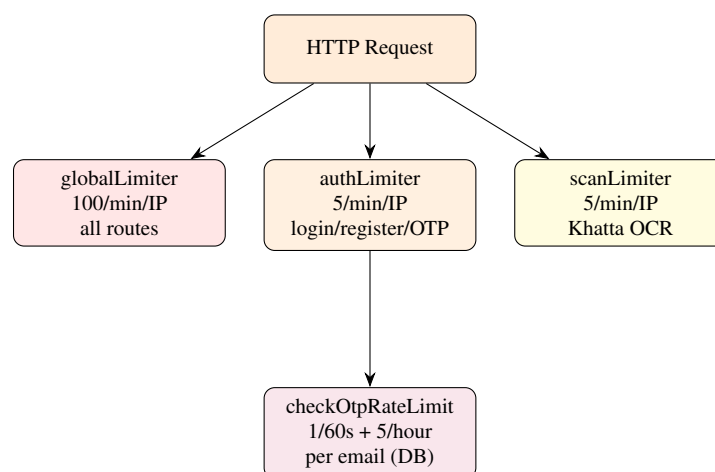


Figure 2.4: Multi-Layer Rate Limiting Architecture



Chapter 3

Database Schema Diagrams

Figure 3.1: Database Schema — Identity and Roles Domain

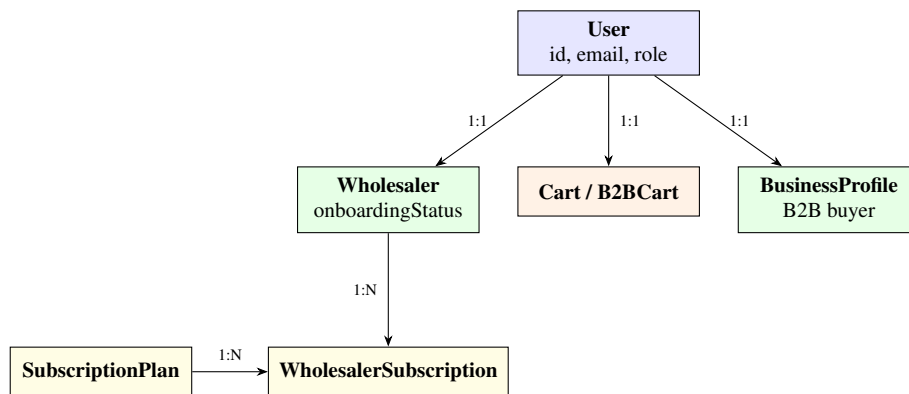


Figure 3.2: Database Schema — Commerce Domain

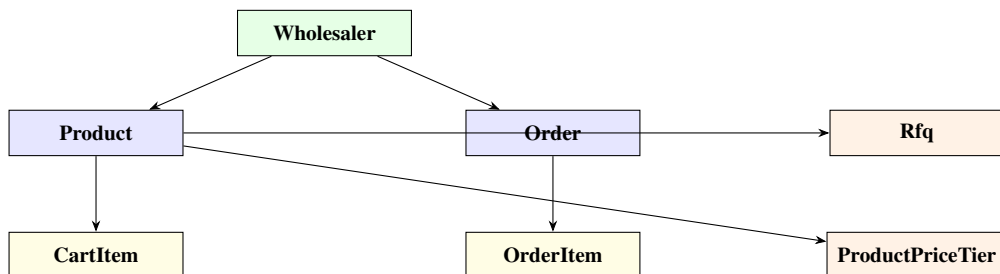


Figure 3.3: Database Schema — Financial and Accounting Domain

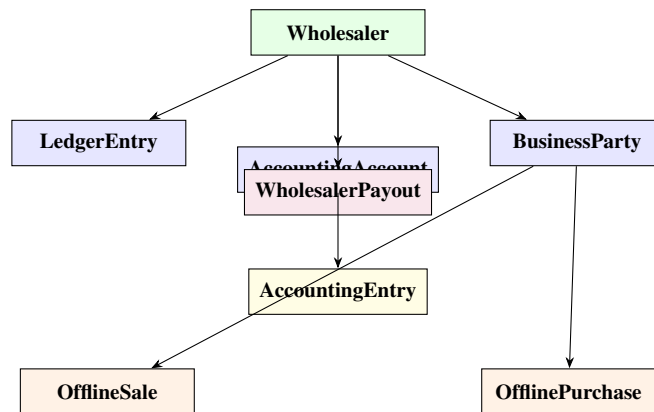


Figure 3.4: Database Schema — Recommendation and AI Attribution Domain

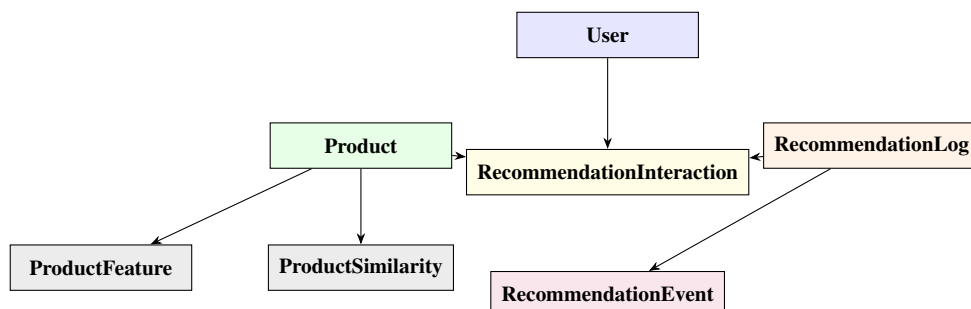
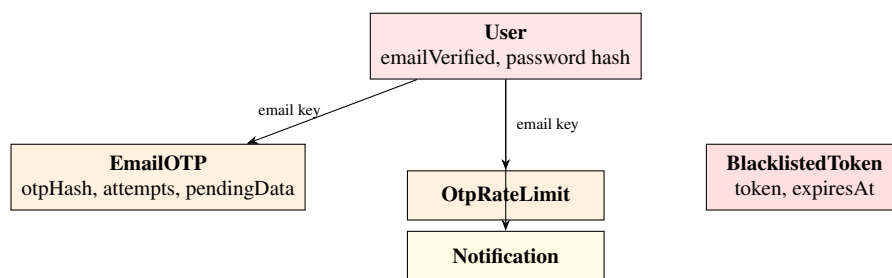


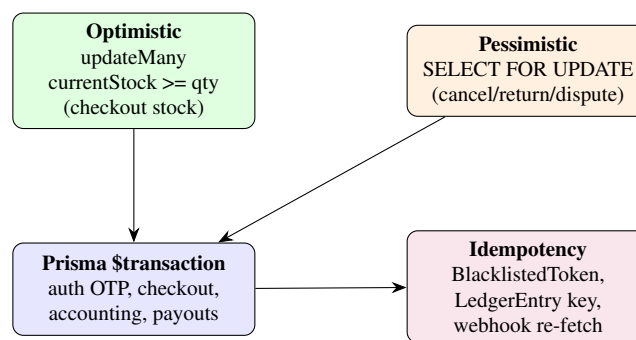
Figure 3.5: Database Schema — Security and Session Domain



Chapter 4

Concurrency and Transaction Patterns

Figure 4.1: Concurrency Control Patterns



Chapter 5

Business Process Workflows

Figure 5.1: B2C Order Checkout Flow (COD and Prepaid)

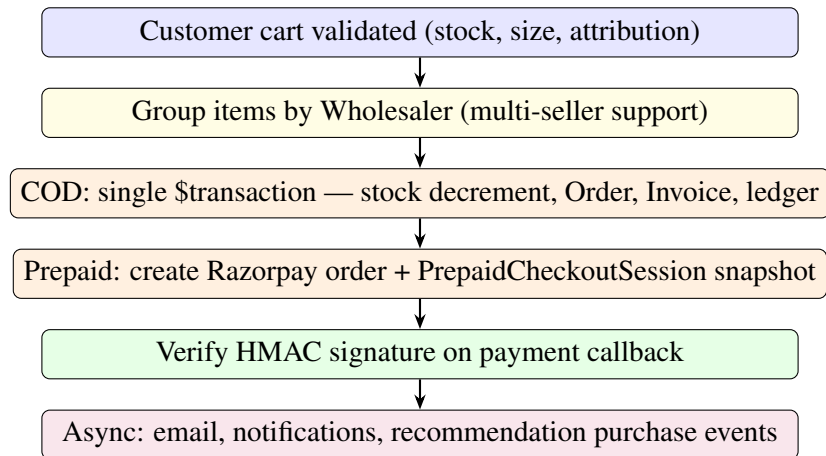
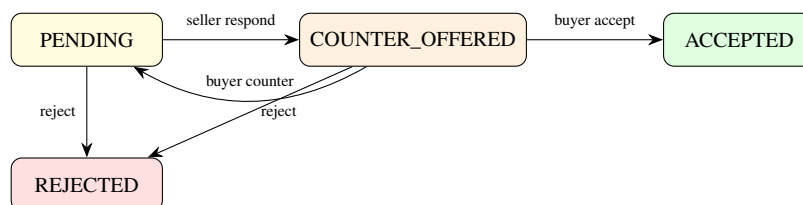


Figure 5.2: B2B RFQ Negotiation Workflow



Chapter 6

AI and Recommendation Systems

Figure 6.1: Hybrid Recommendation Score Fusion Pipeline

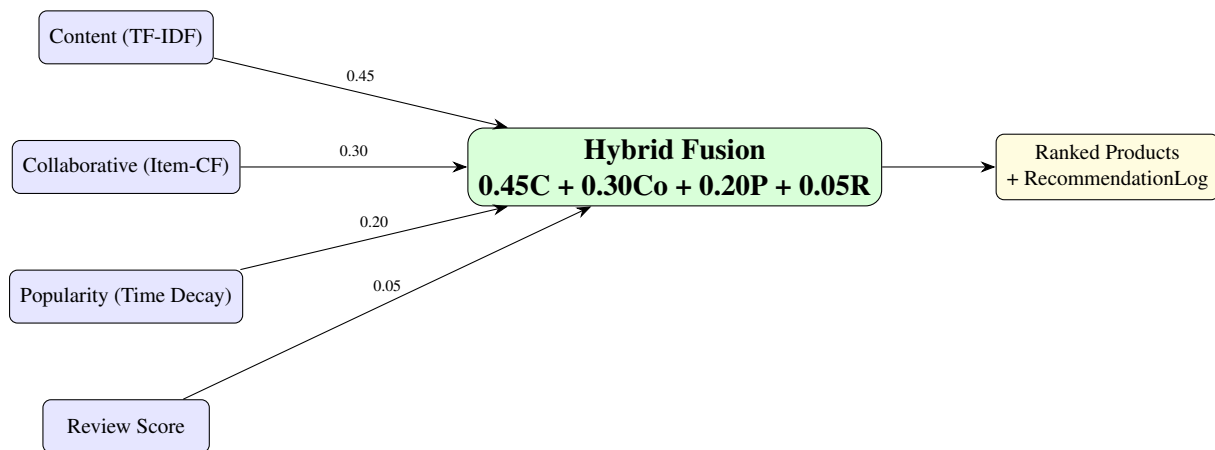


Figure 6.2: Offline Recommendation Job Pipeline

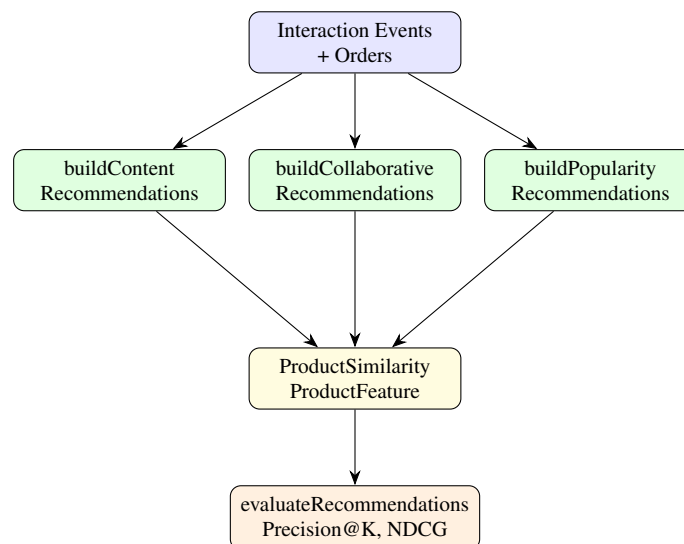


Figure 6.3: AI Business Advisor RAG Pipeline

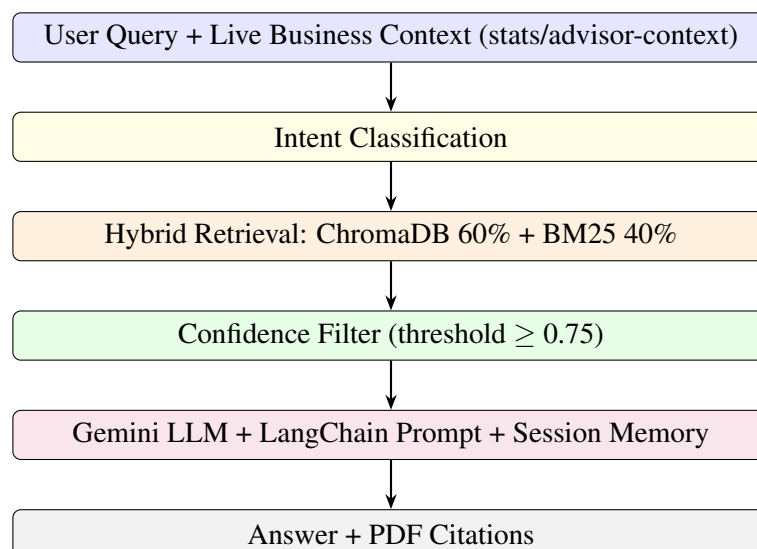


Figure 6.4: AI Khatta Vision OCR Digitisation Flow

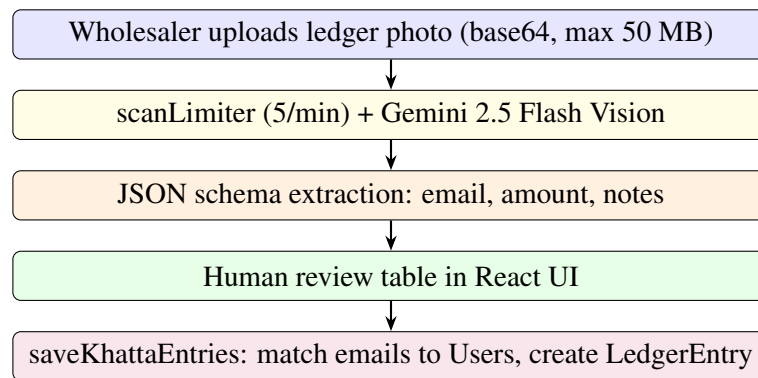
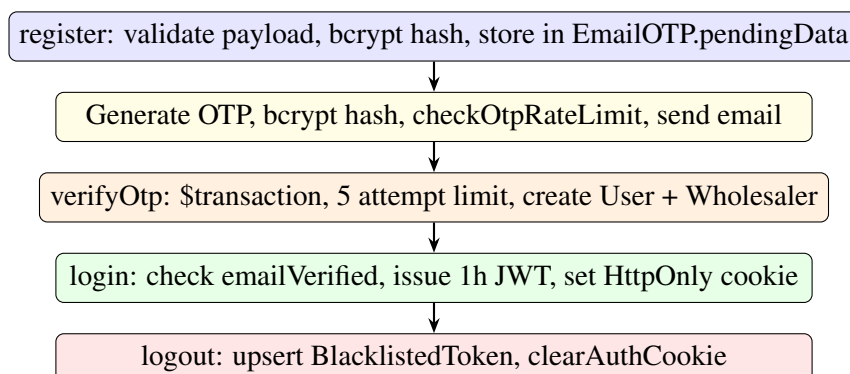


Figure 6.5: Two-Step OTP Registration and Login Security Flow



Chapter 7

Frontend Architecture

Figure 7.1: Frontend Application Architecture

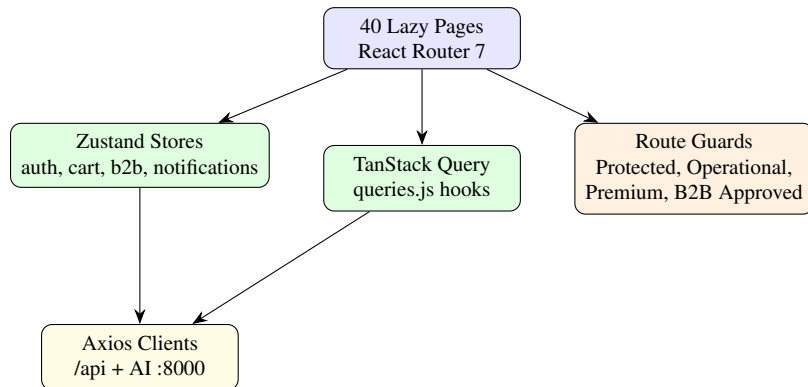


Figure 7.2: Role-Based Access Control Matrix

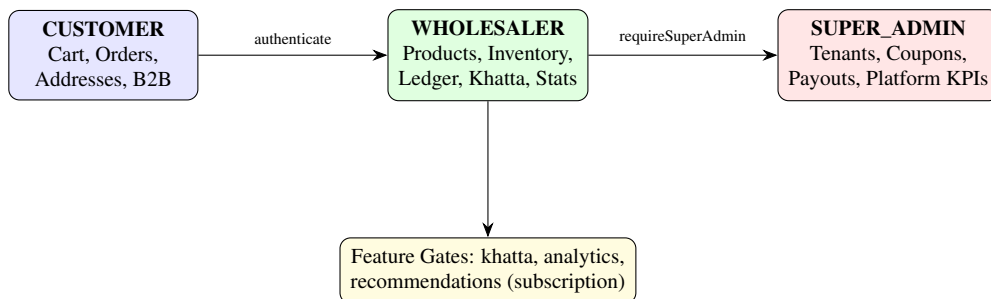


Figure 7.3: Razorpay Webhook Security and Idempotent Processing

